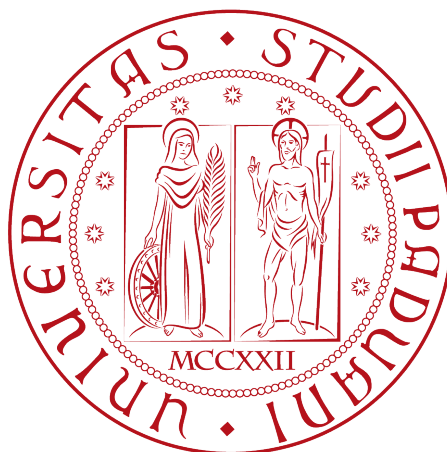


University of Padua

DEPARTMENT OF MATHEMATICS “TULLIO LEVI-CIVITA”

MASTER DEGREE IN COMPUTER SCIENCE



GeoMedia: A Geographic Framework for Location-Based Content Sharing

Master's Thesis

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Summary

Over the past few decades, the use of mobile devices has increased drastically, almost replacing desktop software and web browsers. This shift has been driven not only by the widespread adoption of smartphones, but also by the availability of integrated sensors such as GPS, gyroscopes, and Bluetooth, which allows a richer and more context-aware user experiences. By leveraging these technologies, Online Social Media platforms have been able to provide more advanced and sophisticated features; consequently collecting larger amounts of metadata, interaction and daily engagement.

This thesis presents “GeoMedia”, an open-source framework designed for sharing multimedia content — such as text, images, audio, and files — anchored to real-world geographic locations. By leveraging GPS data available on mobile devices, the application provides an interactive map through which users can explore and publish digital content, creating a seamless integration between the digital and physical worlds. Furthermore, the framework offers several features and cover various use cases, including restricting content visibility within customizable geographic areas, enabling time-sensitive content, applying user-based restrictions for content creation and visualization, and supporting sequential visibility between published posts. Furthermore, analyzes the motivations behind GeoMedia, its critical aspects and design considerations, describes its system architecture and implementation through a comparison of different technological paradigms. Finally, it explores potential applications and outlines possible future developments of the framework.

An previous version of this framework has been presented at “GoodIT ’25: International Conference on Information Technology for Social Good” (September 3 - 5, 2025 - Antwerp, Belgium) [1]

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Chapter 1

Introduction

1.1 Overview

Over the last two decades, mobile devices such as smartphones, tablets, and wearable technologies have become deeply integrated into everyday life, shaping people's daily habits and their interaction with technology and digital applications, almost replacing traditional desktop software and web browsing. The adoption of smartphones has been driven by several factors, the growing computational power, battery increasing and wider functionalities compared to conventional computers due to their built-in sensors - such as gyroscopes, GPS modules, light sensors, cameras and vibration motors. The presence of these features have allowed application to become create a more immersive user experience and provide different integration in different fields like health, education and physical world - through sensors or cameras -

A lot of examples of this trend can be observed in modern society. For instance, when creating an itinerary or even while driving, smartphones assist users through applications such as Google Maps and Waze. These applications do not merely provide directions, but also enhance the travel experience by offering real-time traffic updates, road event notifications, and real-time information about available parking lots. In this way, the surrounding environment has been enriched by digital detail, another notable example is Google Lens, which allows to identify objects from a photograph, or Shazam that can recognize a song through the device's microphone. The technology integration has become a crucial key even in work environments, from the restaurant that provide a digital menu by a tablet, to customers consulting shopping lists or mall stores. Similarly, many museums allowing visitors to scan QR codes to access audio-guided tours or unlock Augmented Reality functionalities that enrich the overall experience.

1.2 Social networks

Social networks such as WhatsApp, Telegram, Instagram, and Facebook have attracted billions of users worldwide, becoming an integral part of modern digital communication. These applications allow users to share diverse types of content—including photos, audio recordings, videos, and text messages—with individuals, groups, or even a global audience. This has expanded the possibilities for communication, collaboration, and the dissemination of information across personal, professional, and public contexts.

These platforms are primarily user-centric, relying on personalized algorithms that suggest content based on users' interests, interactions, and browsing habits. By facilitating the creation and maintenance of digital connections, social media has fostered increasingly interconnected social networks, bridging geographical distances and enabling communities

to form around shared interests. Beyond communication, these platforms have also become channels for entertainment, news consumption, and even professional networking, reflecting their multifaceted role in contemporary society.

SOCIAL MEDIA

- snapchat - instagram - google maps / trip advisor - foursquare - jagat - auglinn - wallame??

Esempio di utilizzo di un termine nel glossario

[Application Program Interface \(API\)](#).

Esempio di citazione in linea

[3].

Esempio di citazione nel pie' di pagina

citazione¹

1.3 L'azienda

Descrizione dell'azienda.

1.4 L'idea

Introduzione all'idea dello stage.

1.5 Organizzazione del testo

Il secondo capitolo descrive ...

Il terzo capitolo approfondisce ...

Il quarto capitolo approfondisce ...

Il quinto capitolo approfondisce ...

Il sesto capitolo approfondisce ...

Nel settimo capitolo descrive ...

```
1 def hello():
2     print("Hello")
```

Listing 1.1: Python example

```
1 function hello() {
2     console.log("Hello");
3 }
```

Listing 1.2: JavaScript example

```
1 SELECT * FROM users;
```

Listing 1.3: SQL example

¹2.

Chapter 2

Related works

2.1 Instagram and Snapchat

2.2 Google Maps

2.3 Strava

2.4 WallaMe

Chapter 3

Client Layer

Brevissima introduzione al capitolo

3.1 Processo sviluppo prodotto

Chapter 4

Server Layer

Brevissima introduzione al capitolo

4.1 Processo sviluppo prodotto

Chapter 5

Data Layer

Brevissima introduzione al capitolo

5.1 Processo sviluppo prodotto

Chapter 6

Conclusions

- 6.1 Consuntivo finale
- 6.2 Raggiungimento degli obiettivi
- 6.3 Conoscenze acquisite
- 6.4 Valutazione personale

“Life is really simple, but we insist on making it complicated”

— Confucius

Ringraziamenti

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Desidero ringraziare con affetto i miei genitori per il sostegno, il grande aiuto e per essermi stati vicini in ogni momento durante gli anni di studio.

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Padua, May 2026

Alberto Morini

Appendix A

Appendice A

Citazione

Autore della citazione

Acronimi e abbreviazioni

API [Application Program Interface](#). [2](#), [11](#)

UML [Unified Modeling Language](#). [11](#)

Glossario

API in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione. [10](#)

UML in ingegneria del software *UML, Unified Modeling Language* (ing. linguaggio di modellazione unificato) è un linguaggio di modellazione e specifica basato sul paradigma object-oriented. L'*UML* svolge un'importantissima funzione di “lingua franca” nella comunità della progettazione e programmazione a oggetti. Gran parte della letteratura di settore usa tale linguaggio per descrivere soluzioni analitiche e progettuali in modo sintetico e comprensibile a un vasto pubblico. [10](#)

Bibliografia

Riferimenti bibliografici

- [2] Daniel T. Jones James P. Womack. *Lean Thinking, Second Editon*. Simon & Schuster, Inc., 2010 (cit. on p. [2](#)).

Siti web consultati

- [1] *GoodIT 2025 - International Conference on Information Technology for Social Good*. 2025. URL: <https://goodit2025.org> (cit. on p. [ii](#)).
- [3] *Manifesto Agile*. URL: <http://agilemanifesto.org/iso/it/> (cit. on p. [2](#)).